# Sip Studies × Sipopedia Beverage Taxonomy

Guild > academy > subgenre > styles. Source data: data/guilds.json. Beer follows BJCP 2021 categories 1–34.

## 1. Cask Guild — Fermentation, transformation, and time (118 subgenres, 740 styles)

Grouped by maturation, cellar craft, and cellar-to-service.

### 1.1 Wine Academy (45 subgenres, 318 styles)

#### 1.1.1 Testable Red
Cabernet Sauvignon · Merlot · Pinot Noir · Syrah/Shiraz · Grenache/Garnacha · Tempranillo · Sangiovese · Nebbiolo · Malbec · Zinfandel/Primitivo · Cabernet Franc · Gamay · Barbera · Mourvèdre/Monastrell

#### 1.1.2 Testable White
Chardonnay · Sauvignon Blanc · Riesling · Pinot Grigio/Gris · Chenin Blanc · Gewürztraminer · Viognier · Albariño · Grüner Veltliner · Sémillon · Torrontés · Assyrtiko

#### 1.1.3 Rosé
Provence rosé · Tavel · Rosado (Navarra) · Rosato · White Zinfandel

#### 1.1.4 Sparkling
Champagne · Crémant · Cava · Prosecco · Franciacorta · Sekt · Pét-nat (ancestral method) · Lambrusco · Moscato d'Asti

#### 1.1.5 Orange (Skin-Contact)
Georgian qvevri (Rkatsiteli) · Friulian Ribolla Gialla · Slovenian (Brda) skin-contact

#### 1.1.6 Fortified
Port (Ruby/Tawny/Vintage) · Sherry (Fino/Manzanilla/Amontillado/Oloroso/PX) · Madeira · Marsala · Vermouth · Vin doux naturel

#### 1.1.7 Dessert
Sauternes · Tokaji Aszú · Icewine/Eiswein · Beerenauslese/TBA · Vin Santo · Recioto · Late harvest

#### 1.1.8 Non-Testable Red › France
Tannat · Petit Verdot · Carignan · Cinsault · Pinot Meunier · Négrette · Mondeuse · Poulsard · Trousseau · Fer Servadou

#### 1.1.9 Non-Testable Red › Italy
Aglianico · Nero d'Avola · Montepulciano · Dolcetto · Corvina · Lagrein · Sagrantino · Nerello Mascalese · Teroldego · Schiava

#### 1.1.10 Non-Testable Red › Spain & Portugal
Mencía · Bobal · Graciano · Prieto Picudo · Touriga Nacional · Touriga Franca · Baga · Castelão

#### 1.1.11 Non-Testable Red › Central & Eastern Europe
Blaufränkisch · Zweigelt · St. Laurent · Kadarka · Plavac Mali · Xinomavro · Agiorgitiko · Fetească Neagră · Mavrud

#### 1.1.12 Non-Testable Red › New World
Carménère (Chile) · País (Chile) · Pinotage (South Africa) · Bonarda (Argentina)

#### 1.1.13 Non-Testable White › France
Marsanne · Roussanne · Melon de Bourgogne · Aligoté · Petit Manseng · Gros Manseng · Savagnin · Picpoul · Clairette · Mauzac

#### 1.1.14 Non-Testable White › Italy
Garganega · Vermentino · Fiano · Greco · Falanghina · Cortese · Arneis · Trebbiano · Verdicchio · Friulano

#### 1.1.15 Non-Testable White › Spain & Portugal
Verdejo · Godello · Macabeo (Viura) · Xarel·lo · Parellada · Palomino · Loureiro · Arinto · Encruzado

#### 1.1.16 Non-Testable White › Central & Eastern Europe
Furmint · Hárslevelű · Welschriesling · Silvaner · Müller-Thurgau · Kerner · Rkatsiteli · Moschofilero · Malagousia

#### 1.1.17 Non-Testable White › Other
Muscat of Alexandria · Colombard · Chasselas · Petite Arvine

#### 1.1.18 Non-Vitis vinifera › Vitis labrusca
Concord · Catawba · Niagara · Isabella · Ives

#### 1.1.19 Non-Vitis vinifera › Vitis riparia & rupestris
Vitis riparia (riverbank grape) · Vitis rupestris (rock grape) · Vitis berlandieri · Mainly rootstock parents — see Rootstocks

**1.1.20 Non-Vitis vinifera › Vitis rotundifolia (Muscadine)**
Scuppernong · Carlos · Noble · Magnolia

**1.1.21 Non-Vitis vinifera › Vitis aestivalis**
Norton (Cynthiana)

**1.1.22 Non-Vitis vinifera › Interspecific Hybrids**
Seyval Blanc · Vidal Blanc · Chambourcin · Baco Noir · Maréchal Foch · Traminette · Marquette · Frontenac · La Crescent

**1.1.23 Non-Vitis vinifera › PIWI (Fungus-Resistant)**
Regent · Solaris · Johanniter · Souvignier Gris · Cabernet Cortis · Muscaris · Divico

**1.1.24 Vine Genetics › Clones**
Pinot Noir Dijon 115 · Pinot Noir Dijon 667 · Pinot Noir Dijon 777 · Pinot Noir Pommard (UCD 4) · Pinot Noir Wädenswil (UCD 1A) · Chardonnay Dijon 76 · Chardonnay Dijon 96 · Chardonnay Wente · Cabernet Sauvignon 337 · Sangiovese Grosso (Brunello)

**1.1.25 Vine Genetics › Mutations (Bud Sports)**
Pinot Gris (from Pinot Noir) · Pinot Blanc · Pinot Meunier (chimera) · Grenache Blanc · Grenache Gris · Gewürztraminer (aromatic Savagnin Rose) · Chardonnay Musqué · Sauvignon Gris

**1.1.26 Vine Genetics › Teinturiers (Red-Fleshed)**
Alicante Bouschet · Saperavi · Petit Bouschet · Grand Noir de la Calmette · Gamay de Bouze · Rubired

**1.1.27 Rootstocks › Riparia & Rupestris Types**
Riparia Gloire de Montpellier · Rupestris du Lot (St. George) · 3309 Couderc · 101-14 Millardet et de Grasset · Schwarzmann

**1.1.28 Rootstocks › Berlandieri × Riparia**
5BB Kober · SO4 · 5C Teleki · 420A Millardet et de Grasset

**1.1.29 Rootstocks › Berlandieri × Rupestris**
110 Richter · 1103 Paulsen · 140 Ruggeri · 99 Richter

**1.1.30 Rootstocks › Nematode-Resistant & Other**
Freedom · Harmony · Dog Ridge · Ramsey (Salt Creek) · 1616 Couderc · Own-rooted (ungrafted)

**1.1.31 Canopy Management › Training Systems**
Single Guyot · Double Guyot · Bilateral cordon (spur-pruned) · Vertical shoot positioning (VSP) · Gobelet (bush vine) · Pergola · Lyre · Scott Henry · Smart-Dyson · Geneva Double Curtain · Sprawl

**1.1.32 Canopy Management › Canopy Practices**
Cane pruning · Spur pruning · Shoot thinning · Shoot positioning · Leaf removal · Hedging (topping) · Green harvest (crop thinning) · Cover crops

**1.1.33 Winemaking Techniques › Red Winemaking**
Destemming · Whole-cluster fermentation · Cold soak · Carbonic maceration · Pump-over (remontage) · Punch-down (pigeage) · Délestage (rack & return) · Extended maceration · Saignée · Malolactic conversion

**1.1.34 Winemaking Techniques › White & Rosé Winemaking**
Whole-bunch pressing · Skin contact · Cold settling (débourbage) · Cool fermentation · Lees aging (sur lie) · Bâtonnage · Direct-press rosé · Blocked malolactic

**1.1.35 Winemaking Techniques › Sparkling Methods**
Traditional method · Tank (Charmat) method · Ancestral method · Transfer method · Riddling (remuage) · Disgorgement · Dosage · Carbonation

**1.1.36 Winemaking Techniques › Fortified & Oxidative**
Mutage (fortification during fermentation) · Post-fermentation fortification · Flor (biological aging) · Oxidative aging · Estufagem (Madeira)

**1.1.37 Winemaking Techniques › Finishing**
Fining · Filtration · Cold stabilization · Blending

**1.1.38 Barrel Toasting & Aging › Oak & Other Woods**
American white oak (Quercus alba) · French sessile oak (Q. petraea) · French pedunculate oak (Q. robur) · Hungarian oak (Q. petraea) · Oregon oak (Q. garryana) · Mizunara (Q. mongolica, Japan) · Acacia (black locust) · Chestnut (Castanea sativa) · Cherry (Prunus avium) · Amburana (Brazil)

**1.1.39 Barrel Toasting & Aging › French Oak Forests**
Allier · Tronçais (Allier) · Nevers · Bertranges · Vosges · Limousin (wide grain, Cognac) · Jupilles

**1.1.40 Barrel Toasting & Aging › Seasoning**
Open-air seasoning (typically 24–36 months) · Kiln-dried staves · Rain & sun leaching of harsh tannins

**1.1.41 Barrel Toasting & Aging › Toast Levels**

Light toast · Medium toast · Medium-plus toast · Heavy toast · Toasted heads · Untoasted heads · See Spirits Academy: char levels

#### 1.1.42 Barrel Toasting & Aging › Barrel Sizes
Blood tub (~40 L) · Quarter cask (~125 L) · American standard barrel (200 L / 53 gal) · Bordeaux barrique (225 L) · Burgundy pièce (228 L) · Hogshead (~250 L) · Puncheon (~450–500 L) · Sherry butt (~500 L) · Port pipe (~550–650 L) · Foudre / foeder (1,000 L+)

#### 1.1.43 Barrel Toasting & Aging › New vs Used
New oak · First fill · Refill (second fill+) · Neutral oak (3+ uses) · Rejuvenated (re-toasted/re-charred) · Percentage new oak (wine élevage)

#### 1.1.44 Wine-Based Drinks
Sangria · Spritzer · Wine cooler · Mulled wine · Kalimotxo (wine + cola)

#### 1.1.45 Dealcoholized Wine (≤0.5% ABV)
Dealcoholized red · Dealcoholized white · Dealcoholized sparkling · Dealcoholized rosé

### 1.2 Beer Academy (46 subgenres, 218 styles)

#### 1.2.1 1 Standard American Beer
American Light Lager · American Lager · Cream Ale · American Wheat Beer

#### 1.2.2 2 International Lager
International Pale Lager · International Amber Lager · International Dark Lager

#### 1.2.3 3 Czech Lager
Czech Pale Lager · Czech Premium Pale Lager · Czech Amber Lager · Czech Dark Lager

#### 1.2.4 4 Pale Malty European Lager
Munich Helles · Festbier · Helles Bock

#### 1.2.5 5 Pale Bitter European Beer
German Leichtbier · Kölsch · German Helles Exportbier · German Pils

#### 1.2.6 6 Amber Malty European Lager
Märzen · Rauchbier · Dunkles Bock

#### 1.2.7 7 Amber Bitter European Beer
Vienna Lager · Altbier

#### 1.2.8 8 Dark European Lager
Munich Dunkel · Schwarzbier

#### 1.2.9 9 Strong European Beer
Doppelbock · Eisbock · Baltic Porter

#### 1.2.10 10 German Wheat Beer
Weissbier · Dunkles Weissbier · Weizenbock

#### 1.2.11 11 British Bitter
Ordinary Bitter · Best Bitter · Strong Bitter

#### 1.2.12 12 Pale Commonwealth Beer
British Golden Ale · Australian Sparkling Ale · English IPA

#### 1.2.13 13 Brown British Beer
Dark Mild · British Brown Ale · English Porter

#### 1.2.14 14 Scottish Ale
Scottish Light · Scottish Heavy · Scottish Export

#### 1.2.15 15 Irish Beer
Irish Red Ale · Irish Stout · Irish Extra Stout

#### 1.2.16 16 Dark British Beer
Sweet Stout · Oatmeal Stout · Tropical Stout · Foreign Extra Stout

#### 1.2.17 17 Strong British Ale
British Strong Ale · Old Ale · Wee Heavy · English Barley Wine

#### 1.2.18 18 Pale American Ale
Blonde Ale · American Pale Ale

#### 1.2.19 19 Amber and Brown American Beer
American Amber Ale · California Common · American Brown Ale

#### 1.2.20 20 American Porter and Stout

American Porter · American Stout · Imperial Stout

### 1.2.21 21 IPA
American IPA · Specialty IPA: Belgian IPA · Specialty IPA: Black IPA · Specialty IPA: Brown IPA · Specialty IPA: Red IPA · Specialty IPA: Rye IPA · Specialty IPA: White IPA · Specialty IPA: Brut IPA · Hazy IPA

### 1.2.22 22 Strong American Ale
Double IPA · American Strong Ale · American Barleywine · Wheatwine

### 1.2.23 23 European Sour Ale
Berliner Weisse · Flanders Red Ale · Oud Bruin · Lambic · Gueuze · Fruit Lambic · Gose

### 1.2.24 24 Belgian Ale
Witbier · Belgian Pale Ale · Bière de Garde

### 1.2.25 25 Strong Belgian Ale
Belgian Blond Ale · Saison · Belgian Golden Strong Ale

### 1.2.26 26 Monastic Ale
Trappist Single · Belgian Dubbel · Belgian Tripel · Belgian Dark Strong Ale

### 1.2.27 27 Historical Beer
Kellerbier · Kentucky Common · Lichtenhainer · London Brown Ale · Piwo Grodziskie · Pre-Prohibition Lager · Pre-Prohibition Porter · Roggenbier · Sahti

### 1.2.28 28 American Wild Ale
Brett Beer · Mixed-Fermentation Sour Beer · Wild Specialty Beer · Straight Sour Beer

### 1.2.29 29 Fruit Beer
Fruit Beer · Fruit and Spice Beer · Specialty Fruit Beer · Grape Ale

### 1.2.30 30 Spiced Beer
Spice, Herb, or Vegetable Beer · Autumn Seasonal Beer · Winter Seasonal Beer · Specialty Spice Beer

### 1.2.31 31 Alternative Fermentables Beer
Alternative Grain Beer · Alternative Sugar Beer

### 1.2.32 32 Smoked Beer
Classic Style Smoked Beer · Specialty Smoked Beer

### 1.2.33 33 Wood Beer
Wood-Aged Beer · Specialty Wood-Aged Beer

### 1.2.34 34 Specialty Beer
Commercial Specialty Beer · Mixed-Style Beer · Experimental Beer

### 1.2.35 Barrel Toasting & Aging › Barrel-Aged & Wood Beers
Bourbon barrel-aged imperial stout · Barrel-aged barleywine · Foeder-fermented beer · Wine-barrel sour · Brettanomyces aged in oak · Lambic in oak foudres · See Wine Academy: oak, toast, sizes

### 1.2.36 Flavored Malt & Hard Seltzer (outside BJCP)
Hard seltzer · Flavored malt beverage · Malt liquor · Hard root beer/soda · Hard tea

### 1.2.37 NA & Low-Alcohol Beer (outside BJCP)
Non-alcoholic lager (≤0.5% ABV) · Non-alcoholic IPA · Non-alcoholic stout · Radler/shandy

### 1.2.38 Cultivation & Raw Materials › Malts
Pilsner malt · Pale ale malt · Maris Otter · Vienna malt · Munich malt · Wheat malt · Rye malt · Crystal/caramel malts · Chocolate malt · Black (patent) malt · Roasted barley · Smoked malt · Acidulated malt

### 1.2.39 Cultivation & Raw Materials › Hops: US & Southern Hemisphere
Cascade · Centennial · Citra · Mosaic · Simcoe · Amarillo · Chinook · Galaxy (Australia) · Nelson Sauvin (NZ) · Motueka (NZ)

### 1.2.40 Cultivation & Raw Materials › Hops: European
Saaz · Hallertauer Mittelfrüh · Tettnanger · Spalter · East Kent Goldings · Fuggle · Styrian Golding

### 1.2.41 Science › Yeast & Microbes
Ale yeast (S. cerevisiae) · Lager yeast (S. pastorianus) · Kveik · Belgian/Saison yeast · Hefeweizen yeast · Brettanomyces · Lactobacillus · Pediococcus

### 1.2.42 Science › Water Profiles
Burton-on-Trent (high sulfate) · Pilsen (very soft) · Munich (carbonate) · Dublin (carbonate) · London · Dortmund · Vienna · Sulfate-to-chloride ratio

### 1.2.43 Production › Brewing Process
Malting (steep, germinate, kiln) · Milling · Mashing · Lautering · Sparging · Boil · Whirlpool · Wort chilling · Fermentation · Lagering/conditioning · Packaging

### 1.2.44 Techniques › Mash Methods
Single infusion · Step infusion · Single decoction · Double decoction · Triple decoction · Cereal mash · Protein rest · Saccharification rest

### 1.2.45 Techniques › Hopping Techniques
Bittering addition · Flavor/aroma addition · Whirlpool hop stand · Dry hopping · Biotransformation · First wort hopping · CO2 hop extract

### 1.2.46 Science › Measurements
Original gravity (OG) · Final gravity (FG) · ABV · IBU · SRM color · EBC color · Apparent attenuation · Degrees Plato

## 1.3 Spirits Academy (27 subgenres, 204 styles)

### 1.3.1 American Whiskey
Bourbon · Straight Bourbon · Bottled-in-Bond · Tennessee Whiskey · Rye Whiskey · Wheat Whiskey · Corn Whiskey · American Single Malt

### 1.3.2 Scotch Whisky
Speyside · Highland · Lowland · Islay · Campbeltown · Islands (Highland sub-region) · Single Malt · Blended Malt · Single Grain · Blended Scotch

### 1.3.3 Irish Whiskey
Single Pot Still · Single Malt · Single Grain · Blended

### 1.3.4 Japanese Whisky
Single Malt · Blended · Grain

### 1.3.5 Other Whisk(e)y
Canadian Whisky · Indian Single Malt · Taiwanese Single Malt · Welsh Whisky

### 1.3.6 Gin
London Dry · Plymouth · Old Tom · Genever (Jenever) · Navy Strength · Contemporary/New Western · Sloe Gin (liqueur)

### 1.3.7 Rum
Light/White · Gold · Dark · Overproof · Spiced · Rhum Agricole (AOC Martinique) · Cachaça · Jamaican pot-still · Demerara (Guyana)

### 1.3.8 Agave
Tequila Blanco · Tequila Joven · Tequila Reposado · Tequila Añejo · Tequila Extra Añejo · Mezcal (Joven/Reposado/Añejo) · Mezcal Ancestral · Mezcal Artesanal · Raicilla · Sotol (Dasylirion) · Bacanora

### 1.3.9 Brandy
Cognac VS · Cognac VSOP · Cognac XO · Armagnac · Brandy de Jerez · Pisco (Peru/Chile) · Calvados · Grappa · Eau-de-vie (Kirsch/Poire Williams) · American brandy

### 1.3.10 Vodka
Grain vodka · Potato vodka · Grape vodka · Flavored vodka

### 1.3.11 Liqueurs
Amaro · Orange liqueur (Triple sec/Curaçao) · Herbal (Chartreuse/Bénédictine) · Anise (Pastis/Ouzo/Sambuca/Absinthe) · Coffee liqueur · Cream liqueur · Nut liqueur (Amaretto/Frangelico) · Fruit liqueur (Limoncello/Crème de cassis)

### 1.3.12 Asian Spirits
Baijiu: Strong aroma (nongxiang) · Baijiu: Light aroma (qingxiang) · Baijiu: Sauce aroma (jiangxiang) · Baijiu: Rice aroma (mixiang) · Shochu (honkaku) · Awamori · Soju · Arrack

### 1.3.13 Barrel Toasting & Aging › Char Levels
#1 char (light) · #2 char · #3 char (common for bourbon) · #4 char (alligator) · Toast + char (two-step) · See Wine Academy: oak, forests, sizes

### 1.3.14 Barrel Toasting & Aging › Cask Finishes
Ex-bourbon · Oloroso sherry · Pedro Ximénez (PX) sherry · Fino/Amontillado sherry · Port (ruby/tawny) · Madeira · Sauternes · Red wine · Rum cask · Cognac cask · Beer/stout cask

### 1.3.15 Barrel Toasting & Aging › Maturation Systems
Solera (criaderas y solera) · Angel's share (evaporation) · Dunnage warehouse · Rickhouse (rackhouse) · Palletized warehouse · Tropical maturation · Bottled-in-Bond (≥4 years, US)

### 1.3.16 Barrel Toasting & Aging › Agave Aging (NOM-006)
Tequila Reposado (2–12 months in oak) · Tequila Añejo (≥1 year, ≤600 L oak) · Tequila Extra Añejo (≥3 years) · Cristalino (aged, then filtered)

### 1.3.17 Other Regional Spirits
Aquavit (akvavit) · Rakı · Arak · Pálinka · Slivovitz (šljivovica) · Singani · Feni

### 1.3.18 Cocktails (IBA) › The Unforgettables
Old Fashioned · Manhattan · Dry Martini · Negroni · Daiquiri · Whiskey Sour · Sidecar · Sazerac · Boulevardier · Martinez

#### 1.3.19 Cocktails (IBA) › Contemporary Classics
Margarita · Mojito · Cosmopolitan · Moscow Mule · Mai Tai · Piña Colada · Bloody Mary · Caipirinha · Mimosa · French 75

#### 1.3.20 Cocktails (IBA) › New Era Drinks
Espresso Martini · Spritz Veneziano · Penicillin · Paper Plane · Tommy's Margarita · Naked and Famous

#### 1.3.21 Zero-Proof
Virgin cocktails (mocktails) · Non-alcoholic gin alternatives · Non-alcoholic whiskey alternatives · Non-alcoholic aperitif/bitter alternatives · Shrub-based drinks

#### 1.3.22 Cultivation & Raw Materials › Raw Materials
Corn · Rye · Malted barley · Wheat · Sugarcane juice · Molasses · Grapes/wine · Apples & pears · Blue agave (A. tequilana Weber) · Potatoes · Sorghum · Rice

#### 1.3.23 Cultivation & Raw Materials › Mash Bills
Bourbon (≥51% corn) · Rye whiskey (≥51% rye) · High-rye bourbon · Wheated bourbon · Single malt (100% malted barley) · Four-grain

#### 1.3.24 Production › Fermentation
Sweet mash · Sour mash (backset) · Wash (distiller's beer) · Open vs closed fermenters · Dunder (Jamaican rum) · Solid-state fermentation (baijiu)

#### 1.3.25 Production › Distillation
Pot still · Column (Coffey) still · Hybrid still · Alembic charentais · Double distillation · Triple distillation · Doubler/thumper · Heads (foreshots) · Hearts · Tails (feints) · Reflux · Proof & ABV

#### 1.3.26 Production › Agave Production
Jimador harvest (coa) · Piñas · Earthen pit oven (mezcal) · Masonry horno · Autoclave · Diffuser (difusor) · Tahona (stone wheel) · Roller mill · Copper & clay pot stills

#### 1.3.27 Techniques › Finishing & Bottling
Proofing (dilution) · Chill filtration · Non-chill filtered · Cask strength · Blending (vatting) · Caramel color (E150a) · See Barrel Toasting & Aging

## 2. Steep Guild — Leaf, seed, culture, and extraction (64 subgenres, 415 styles)

Grouped by extraction, infusion, sensory calibration, and ritual.

### 2.1 Coffee Academy (30 subgenres, 208 styles)

#### 2.1.1 Species
Arabica (Coffea arabica) · Robusta (Coffea canephora) · Liberica · Excelsa

#### 2.1.2 Processing
Washed (wet) · Natural (dry) · Honey (pulped natural) · Wet-hulled (Giling Basah) · Anaerobic fermentation · Carbonic maceration

#### 2.1.3 Roast Level
Light (cinnamon/New England) · Medium (City/American) · Medium-dark (Full City) · Dark (Vienna/French/Italian)

#### 2.1.4 Espresso Drinks
Espresso · Ristretto · Lungo · Americano · Macchiato · Cortado · Flat white · Cappuccino · Latte · Mocha · Affogato

#### 2.1.5 Brew Methods
Pour-over (V60/Chemex/Kalita) · French press · AeroPress · Moka pot · Siphon · Drip (batch brew) · Turkish (cezve) · Cold brew · Cold drip (Kyoto) · Phin (Vietnamese)

#### 2.1.6 Barrel Toasting & Aging › Barrel-Aged Coffee
Green beans rested in bourbon barrels · Rum barrel-aged beans · Wine barrel-aged beans · Whiskey barrel-aged cold brew

#### 2.1.7 Decaffeination
Swiss Water Process · CO2 (supercritical) process · Ethyl acetate (sugarcane) process · Methylene chloride (direct/indirect) process

#### 2.1.8 Coffee Formats
Whole bean · Ground · Instant (freeze-dried) · Instant (spray-dried) · Single-serve pods/capsules · Coffee concentrate · Drip bags

#### 2.1.9 Iced & Blended Drinks
Iced coffee · Iced latte · Nitro cold brew · Frappé (Greek) · Blended iced coffee (frappuccino-style) · Espresso tonic · Dalgona coffee · Shakerato

#### 2.1.10 Cultivation & Raw Materials › Arabica Varieties

Typica · Bourbon · Caturra · Catuaí · Mundo Novo · Geisha (Gesha) · SL28 · SL34 · Pacamara · Maragogype · Villa Sarchi · Pink Bourbon

### 2.1.11 Cultivation & Raw Materials › Hybrids & Resistant Cultivars
Timor Hybrid · Catimor · Sarchimor · Castillo · Colombia · Ruiru 11 · Batian · Marsellesa · Obatã · Starmaya (F1 hybrid)

### 2.1.12 Cultivation & Raw Materials › Ethiopian Landraces
Ethiopian heirloom landraces · JARC 74110 · JARC 74112 · Kurume · Wolisho

### 2.1.13 Cultivation & Raw Materials › Robusta & Other Species
Robusta (C. canephora) clones · Conilon (Brazil) · Liberica · Excelsa (C. liberica var. dewevrei)

### 2.1.14 Cultivation & Raw Materials › Growing Conditions
Altitude (masl) · Shade-grown · Full sun · Agroforestry · Volcanic soils · Distinct dry season · Coffee belt (tropics)

### 2.1.15 Cultivation & Raw Materials › Origins: Africa & Arabia
Ethiopia (Yirgacheffe, Sidama, Guji) · Kenya (Nyeri, Kirinyaga) · Rwanda · Burundi · Uganda · Tanzania · Yemen

### 2.1.16 Cultivation & Raw Materials › Origins: Latin America
Colombia · Brazil · Guatemala · Costa Rica · Honduras · El Salvador · Panama · Peru · Mexico · Nicaragua · Bolivia

### 2.1.17 Cultivation & Raw Materials › Origins: Asia-Pacific
Sumatra · Java · Sulawesi · Papua New Guinea · India (Monsooned Malabar) · Vietnam · Yunnan · Hawaii (Kona)

### 2.1.18 Production › Harvest
Selective hand-picking · Strip picking · Mechanical harvesting · Ripeness sorting · Float sorting

### 2.1.19 Production › Wet Processing Steps
Pulping · Fermentation (dry or underwater) · Washing · Mechanical demucilaging · Drying

### 2.1.20 Production › Honey Grades
White honey · Yellow honey · Red honey · Black honey

### 2.1.21 Production › Experimental Processing
Anaerobic natural · Anaerobic washed · Carbonic maceration · Lactic process · Yeast-inoculated · Thermal shock · Co-fermented (infused) · Double fermentation

### 2.1.22 Production › Drying
Patio drying · Raised (African) beds · Mechanical dryers (guardiolas) · Target moisture 10–12% · Resting (reposo)

### 2.1.23 Quality & Standards › Milling & Grading
Hulling (parchment removal) · Polishing · Density sorting · Optical color sorting · Screen size grading · SCA specialty grade · Primary defects (black, sour, fungus) · Secondary defects (broken, insect, quakers)

### 2.1.24 Science › Roast Chemistry
Drying phase · Maillard reactions · Caramelization · Strecker degradation · Chlorogenic acid breakdown · CO2 degassing

### 2.1.25 Techniques › Roast Profile
Charge temperature · Turning point · Rate of rise (RoR) · First crack · Development time ratio · Second crack · Agtron color

### 2.1.26 Science › Extraction & Strength
Extraction yield (18–22%) · Total dissolved solids (TDS) · SCA brewing control chart · Brew ratio (~1:15–1:18) · Refractometer

### 2.1.27 Techniques › Brewing Variables
Grind size · Water temperature (90–96 °C) · Contact time · Agitation · Bypass · Espresso ratio (~1:2) · Pressure (~9 bar)

### 2.1.28 Science › Water for Coffee
SCA water standard (TDS ~150 mg/L) · Calcium hardness · Alkalinity (~40 mg/L) · pH ~7 · Chlorine-free

### 2.1.29 Quality & Standards › Cupping Protocol
SCA cupping form · Fragrance/aroma · Flavor · Aftertaste · Acidity · Body · Balance · Uniformity · Clean cup · Sweetness · Overall · Specialty = 80+ points

### 2.1.30 Quality & Standards › Flavor Wheel (SCA/WCR)
Fruity · Floral · Sweet · Nutty/cocoa · Spices · Roasted · Green/vegetative · Sour/fermented · Other (papery, chemical)

## 2.2 Tea Academy **(24 subgenres, 154 styles)**

### 2.2.1 White Tea
Silver Needle (Bai Hao Yin Zhen) · White Peony (Bai Mu Dan) · Shou Mei · Darjeeling white

### 2.2.2 Yellow Tea
Junshan Yinzhen · Meng Ding Huang Ya · Huoshan Huang Ya

### 2.2.3 Green Tea
Longjing (Dragon Well) · Bi Luo Chun · Gunpowder · Sencha · Gyokuro · Matcha · Hojicha · Genmaicha · Bancha

### 2.2.4 Oolong Tea
Tieguanyin · Da Hong Pao (Wuyi rock) · Dong Ding · Alishan high-mountain · Oriental Beauty · Dan Cong

### 2.2.5 Black Tea
Assam · Darjeeling · Ceylon · Keemun · Dian Hong (Yunnan) · Lapsang Souchong · English Breakfast blend · Earl Grey

### 2.2.6 Dark Tea
Sheng (raw) pu-erh · Shou (ripe) pu-erh · Liu Bao · Fu Zhuan (Anhua brick)

### 2.2.7 Herbal Tisanes
Rooibos · Honeybush · Chamomile · Peppermint · Hibiscus · Ginger · Lemongrass

### 2.2.8 Scented & Blended
Jasmine tea · Earl Grey (bergamot) · Lychee black · Rose black · Osmanthus oolong · Chai spice blend

### 2.2.9 Tea Drinks
Iced tea · Sweet tea (US South) · Milk tea · Chai latte · Matcha latte · London fog · Arnold Palmer (tea + lemonade)

### 2.2.10 Tea Formats
Loose leaf · Tea bags · Pyramid sachets · Compressed cakes/bricks · Powdered (matcha/instant) · RTD bottled tea

### 2.2.11 Cultivation & Raw Materials › Tea Plant Varieties
Camellia sinensis var. sinensis · Camellia sinensis var. assamica · Cambod (hybrid type)

### 2.2.12 Cultivation & Raw Materials › Notable Cultivars
Yabukita · Okumidori · Saemidori · Asatsuyu · Fuding Da Bai · Tieguanyin · Jin Xuan (TTES #12) · Qing Xin · Rou Gui · Shui Xian · Longjing #43 · AV2 (Darjeeling clone)

### 2.2.13 Cultivation & Raw Materials › Growing Regions
China (Fujian, Yunnan, Zhejiang, Anhui) · India (Darjeeling, Assam, Nilgiri) · Sri Lanka (Nuwara Eliya, Uva, Dimbula) · Japan (Shizuoka, Uji, Kagoshima) · Taiwan (Alishan, Nantou) · Kenya · Nepal (Ilam) · Vietnam

### 2.2.14 Cultivation & Raw Materials › Growing Practices
Elevation · Shading (kabuse, tencha) · Seasonal flushes (first, second, autumn) · Plucking table pruning · Organic/biodynamic · Old-tree (gushu) gardens

### 2.2.15 Cultivation & Raw Materials › Plucking Standards
Bud only · One bud, one leaf · One bud, two leaves · One bud, three leaves · Mature leaves (oolong, dark tea)

### 2.2.16 Production › Processing Steps
Withering · Bruising/shaking (oolong) · Rolling · Oxidation · Kill-green (sha qing) · Drying/firing · Sorting · Roasting

### 2.2.17 Production › Kill-Green Methods
Pan-firing (wok) · Steaming (Japan) · Baking · Sun-drying (pu-erh maocha)

### 2.2.18 Production › Process by Tea Type
White: wither + dry · Yellow: kill-green + yellowing (men huang) · Green: kill-green + roll + dry · Oolong: partial oxidation · Black: full oxidation · Dark: heaping (wo dui) / aging

### 2.2.19 Production › Post-Processing
Compression (cakes, bricks) · Aging · Re-roasting · Scenting (jasmine) · Blending

### 2.2.20 Production › Japanese Processing
Aracha (crude tea) · Asamushi (light steam) · Fukamushi (deep steam) · Tencha (for matcha) · Stone-milling · Gyokuro shading (~3 weeks) · Hojicha roasting · Kukicha (stems) · Shiage (refining)

### 2.2.21 Production › Orthodox vs CTC
Orthodox rolling · CTC (crush, tear, curl) · Rotorvane · Fannings · Dust

### 2.2.22 Quality & Standards › Leaf Grades
OP (Orange Pekoe) · FOP · GFOP · TGFOP · FTGFOP · SFTGFOP · BOP (Broken OP) · Fannings · Dust · CTC grades (BP, PD)

### 2.2.23 Techniques › Brewing Methods
Western teapot · Gongfu cha (gaiwan, Yixing) · Grandpa style · Kyusu (Japanese) · Cold brew · Matcha whisking (chasen)

### 2.2.24 Techniques › Brewing Parameters
White/green ~70–85 °C · Oolong ~85–95 °C · Black/dark ~95–100 °C · Western ~2–3 g per 250 mL · Gongfu ~5–7 g per 100 mL · Multiple infusions · Soft water

## 2.3 Kombucha Academy **(10 subgenres, 53 styles)**

### 2.3.1 Culture & Fermentation
SCOBY (bacterial cellulose pellicle) · Starter liquid (backslopping) · Primary fermentation (F1) · Secondary fermentation (F2, bottle conditioning)

### 2.3.2 Base Teas

Black tea kombucha · Green tea kombucha · Oolong kombucha · White tea kombucha · Herbal/rooibos-base kombucha

#### 2.3.3 Kombucha Styles
Original/plain · Fruit-flavored · Ginger · Botanical/herbal · Hard kombucha (>0.5% ABV)

#### 2.3.4 Related Cultures
Jun (green tea and honey) · Kombucha vinegar

#### 2.3.5 Science › SCOBY Microbes
Komagataeibacter xylinus · Acetobacter · Gluconobacter · Brettanomyces (Dekkera) · Zygosaccharomyces · Saccharomyces · Schizosaccharomyces · Lactobacillus (variable)

#### 2.3.6 Science › SCOBY Biology
Bacterial cellulose pellicle · Yeast invertase splits sucrose · Yeast make ethanol · Bacteria oxidize ethanol to acetic acid · Gluconic & glucuronic acids

#### 2.3.7 Production › Brewing Process
Sweet tea base (~50–100 g sugar/L) · Starter liquid (~10–20%) · Initial pH ≤4.6 · F1: 7–14 days at ~20–29 °C · Finished pH ~2.5–3.5 · F2: flavor & bottle · Refrigerate

#### 2.3.8 Quality & Standards › Alcohol Control
Below 0.5% ABV = non-alcoholic (US TTB) · Hard kombucha ≥0.5% (TTB-regulated) · ABV lab testing · Cold chain · Yeast management

#### 2.3.9 Techniques › Flavoring & Carbonation
Fruit juice · Whole fruit · Herbs & botanicals · Ginger · Bottle conditioning · Forced carbonation (kegging)

#### 2.3.10 Quality & Standards › Quality & Safety
Mold (fuzzy, dry spots) · Kahm yeast · Fruit flies · Vinegar eels · Over-acidification · Food-safe vessels (no leaded glaze)

## 3. Source Guild — Origin, nourishment, and custody (41 subgenres, 267 styles)

Grouped by foundational ingredients, safe processing, freshness, and custody.

### 3.1 Water Academy (11 subgenres, 54 styles)

#### 3.1.1 Spring Water
Natural spring · Artesian spring · Glacier/icecap water

#### 3.1.2 Mineral Water
Low-mineral (<500 mg/L TDS) · Medium-mineral · High-mineral (>1500 mg/L TDS) · Naturally carbonated mineral

#### 3.1.3 Sparkling Water
Carbonated (added CO2) · Seltzer

#### 3.1.4 Purified Water
Distilled · Reverse osmosis · Deionized

#### 3.1.5 Functional Water
Alkaline (ionized) · Electrolyte-enhanced · Flavored/infused · Hydrogen water

#### 3.1.6 Tap & Treated
Municipal tap water · Private well water · Carbon-filtered · Boiled (point-of-use) · UV-treated

#### 3.1.7 Cultivation & Raw Materials › Sources & Hydrogeology
Confined aquifers · Unconfined aquifers · Springs · Artesian wells · Surface water (lakes, rivers) · Glacial melt · Desalination

#### 3.1.8 Production › Treatment
Coagulation & flocculation · Sedimentation · Sand filtration · Activated carbon · Reverse osmosis · Ion exchange (softening) · UV disinfection · Ozonation · Chlorination/chloramination · Remineralization

#### 3.1.9 Science › Water Chemistry
TDS (mg/L) · Hardness (Ca, Mg) · Alkalinity (bicarbonate) · pH · Sodium · Silica · Nitrate · Conductivity

#### 3.1.10 Quality & Standards › Standards
US EPA (Safe Drinking Water Act) · US FDA bottled water (21 CFR 165.110) · EU Natural Mineral Water Directive · WHO drinking-water guidelines

#### 3.1.11 Techniques › Water for Beverages
SCA coffee water standard · Tea (soft, low-mineral) · Brewing water profiles (see Beer Academy) · Ice quality

### 3.2 Juice Academy (11 subgenres, 76 styles)

#### 3.2.1 Product Types (FDA)

100% juice · Juice from concentrate · Nectar · Juice drink/cocktail · Smoothie · Lemonade · Limeade · Fruit punch

#### 3.2.2 Processing
Not-from-concentrate (NFC) · Fresh-squeezed (unpasteurized, warning label) · Flash-pasteurized · Cold-pressed (HPP) · Frozen concentrate

#### 3.2.3 Citrus
Orange · Grapefruit · Lemon · Lime · Tangerine/mandarin · Blood orange

#### 3.2.4 Pome & Stone Fruit
Apple (cloudy/clear) · Pear · Tart cherry · Peach nectar · Apricot nectar

#### 3.2.5 Berry & Grape
Concord grape · Cranberry · Pomegranate · Blueberry · Açaí · Elderberry

#### 3.2.6 Tropical
Pineapple · Mango · Passion fruit · Guava · Papaya · Coconut water

#### 3.2.7 Vegetable
Tomato · Carrot · Beet · Celery · Green juice (kale/spinach/cucumber) · Mixed vegetable juice

#### 3.2.8 Cultivation & Raw Materials › Juice Fruit Varieties
Valencia orange · Hamlin orange · Pera orange (Brazil) · Gala apple · Fuji apple · Honeycrisp apple · Concord grape · Niagara grape · Wonderful pomegranate · Kent mango · Alphonso mango · MD2 pineapple

#### 3.2.9 Production › Extraction
Washing & sorting · Citrus extractors (JBT/FMC, Brown) · Hydraulic press · Belt press · Rack-and-cloth press · Decanter centrifuge · Enzymatic maceration (pectinase) · Cold-press

#### 3.2.10 Production › Processing
Clarification & fining · Deaeration · HTST pasteurization · Hot-fill · HPP (~600 MPa) · Evaporation (concentrate) · Aroma recovery · Aseptic storage

#### 3.2.11 Quality & Standards › Quality & Regulation
Brix · Brix/acid ratio · FDA Juice HACCP (5-log reduction) · Unpasteurized warning label · Patulin limits (apple) · % juice labeling (21 CFR 101.30)

### 3.3 Milk Academy **(19 subgenres, 137 styles)**

#### 3.3.1 Fat Level: Whole, 2%, 1%, Skim
Whole milk (≥3.25% milkfat) · Reduced-fat milk (2%) · Low-fat milk (1%) · Fat-free/skim milk (<0.5%)

Standards: Whole milk: 21 CFR 131.110

#### 3.3.2 Cream & Cream Products › Fluid Creams
Half-and-half (10.5–18% fat) · Light (coffee) cream (18–30%) · Light whipping cream (30–36%) · Heavy cream (≥36%) · Whipping cream (30–36%, not std.) · Whipped cream (aerosol)

Standards: Half-and-half: §131.180; Light (coffee) cream: §131.155; Light whipping cream: §131.157; Heavy cream: §131.150

#### 3.3.3 Cream & Cream Products › Cultured Creams
Sour cream · Acidified sour cream · Sour half-and-half · Acidified sour half-and-half · Crème fraîche

Standards: Sour cream: §131.160; Acidified sour cream: §131.162; Sour half-and-half: §131.185; Acidified sour half-and-half: §131.187

#### 3.3.4 Cream & Cream Products › International & Dry Creams
Single cream (UK, ≥18%) · Double cream (UK, ≥48%) · Clotted cream (UK, ≥55%) · Dry cream

Standards: Dry cream: §131.149

#### 3.3.5 Concentrated & Dry
Evaporated milk · Sweetened condensed milk · Concentrated milk · Nonfat dry milk · Dry whole milk

Standards: Evaporated milk: §131.130; Sweetened condensed milk: §131.120; Concentrated milk: §131.115; Nonfat dry milk: §131.125; Dry whole milk: §131.147

#### 3.3.6 Cultured & Acidified Milks
Cultured buttermilk · Acidified milk · Drinkable yogurt (yogurt base) · See Fermented Academy: kefir, kumis · See Regional Academy: lassi, ayran, doogh

Standards: Cultured buttermilk: §131.112; Acidified milk: §131.111; Drinkable yogurt: §131.200

#### 3.3.7 Flavored Milks
Chocolate milk · Strawberry milk · Vanilla milk · Banana milk · Coffee milk · Eggnog · Milkshake

Standards: Eggnog: §131.170

#### 3.3.8 Processing
Raw (unpasteurized) · Pasteurized (HTST) · Vat (low-temperature) pasteurized · Ultra-pasteurized · UHT (aseptic, shelf-stable) · Homogenized · Non-homogenized (cream-top) · Lactose-free · A2 milk · Ultra-filtered (high-protein) · Organic (USDA NOP) · Grass-fed · Vitamin A & D fortified

#### 3.3.9 By Species
Cow · Goat · Sheep · Water buffalo · Camel · Yak · Donkey · Mare

#### 3.3.10 Plant Milks
Oat · Almond · Soy · Rice · Coconut · Cashew · Pea · Hemp · Macadamia · Flax · Hazelnut · Pistachio

#### 3.3.11 Creamers
Dairy coffee creamer (half-and-half based) · Non-dairy creamer (liquid) · Non-dairy creamer (powder) · Oat creamer · Almond creamer · Coconut creamer

#### 3.3.12 Cultivation & Raw Materials › Dairy Cattle Breeds
Holstein-Friesian · Jersey · Guernsey · Brown Swiss · Ayrshire · Milking Shorthorn · Normande · Montbéliarde · Gir · Sahiwal

#### 3.3.13 Cultivation & Raw Materials › Goat & Sheep Breeds
Saanen · Alpine · Nubian · LaMancha · Nigerian Dwarf · East Friesian (sheep) · Lacaune (sheep) · Awassi (sheep)

#### 3.3.14 Cultivation & Raw Materials › Farming & Feeding
Pasture-based (grass-fed) · Total mixed ration (TMR) · Silage · Hay · Free-stall housing · USDA organic (≥120 grazing days)

#### 3.3.15 Production › Milking
Herringbone parlor · Parallel parlor · Rotary parlor · Robotic milking (AMS) · Teat dipping · Somatic cell count (SCC) · Bulk tank cooling (≤7 °C / 45 °F)

#### 3.3.16 Science › Milk Components
Water (~87%) · Fat (~3.5–4%) · Protein (~3.3%) · Casein (~80% of protein) · Whey proteins (~20%) · Lactose (~4.8%) · Minerals (Ca, P) · A1 vs A2 β-casein · Fat globules

#### 3.3.17 Production › Processing Steps
Receiving & testing · Clarification · Separation (cream/skim) · Standardization · Homogenization · HTST (72 °C / 15 s) · Ultra-pasteurization (≥138 °C / 2 s) · UHT aseptic · ESL (extended shelf life) · Microfiltration

#### 3.3.18 Quality & Standards › Grading & Regulation
Grade A (Pasteurized Milk Ordinance) · Manufacturing grade (Grade B) · Antibiotic residue testing · Bacteria count limits · Raw milk sales (state law)

#### 3.3.19 Production › Plant Milk Production
Soaking · Wet milling · Enzymatic hydrolysis (oat) · Filtering/decanting · Fortification (Ca, D, B12) · Homogenization · UHT

## 4. Energy Guild — Evidence, function, and stewardship (26 subgenres, 131 styles)

Grouped by health, supplements, protein systems, and energy formulations.

### 4.1 Health Academy (9 subgenres, 44 styles)

#### 4.1.1 Hydration & Electrolytes
Oral rehydration solution (WHO ORS) · Isotonic sports drinks · Hypotonic electrolyte drinks · Electrolyte powders/tablets

#### 4.1.2 Gut Health
Probiotic shots · Prebiotic/fiber drinks · Apple cider vinegar drinks

#### 4.1.3 Functional Wellness
Vitamin-enhanced water · Collagen drinks · Ginger/turmeric shots · Golden milk (turmeric latte) · Mushroom elixirs (reishi/chaga)

#### 4.1.4 Relaxation & Sleep
Magnesium drinks · L-theanine drinks · Melatonin drinks · Adaptogenic drinks (ashwagandha)

#### 4.1.5 Medical Nutrition
Oral nutritional supplements · Diabetes-specific formulas · Infant formula · Thickened liquids (IDDSI levels 1–4)

#### 4.1.6 Science › Electrolyte Science
Sodium · Potassium · Chloride · Magnesium · Calcium · Osmolality (hypo/iso/hypertonic) · Glucose-sodium co-transport (SGLT1)

#### 4.1.7 Cultivation & Raw Materials › Functional Ingredients
Probiotic strains (Lactobacillus, Bifidobacterium) · Prebiotic fibers (inulin, FOS) · Collagen peptides · Ashwagandha · Curcumin (turmeric) · Ginger · Acetic acid (vinegar)

#### 4.1.8 Quality & Standards › Regulation
Conventional food vs dietary supplement (DSHEA) · Structure/function claims · FDA health claims · Supplement Facts vs Nutrition Facts · Medical foods

#### 4.1.9 Science › Evidence Levels

Meta-analyses & systematic reviews · Randomized controlled trials · Observational studies · Mechanistic & animal studies · Expert opinion

### 4.2 Protein Academy (9 subgenres, 47 styles)

#### 4.2.1 Dairy Proteins
Whey protein concentrate · Whey protein isolate · Hydrolyzed whey · Micellar casein · Milk protein isolate

#### 4.2.2 Plant Proteins
Pea · Soy isolate · Brown rice · Hemp · Pumpkin seed · Blended plant protein

#### 4.2.3 Other Proteins
Egg white protein · Collagen peptides · Beef protein isolate

#### 4.2.4 Formats
Powder (mix-in) · Ready-to-drink (RTD) shake · Clear whey isolate drinks · Protein coffee · High-protein ultra-filtered milk · Meal replacement shake

#### 4.2.5 Use Cases
Post-workout recovery · Mass gainer · Weight management · Overnight (casein)

#### 4.2.6 Production › Protein Processing
Microfiltration · Ultrafiltration · Ion exchange · Enzymatic hydrolysis · Spray drying · Instantizing (lecithin) · Concentrate (WPC 34–80%) · Isolate (≥90% protein)

#### 4.2.7 Science › Protein Quality Science
PDCAAS · DIAAS · Leucine content · Essential amino acids · Fast (whey) vs slow (casein) digestion · Branched-chain amino acids

#### 4.2.8 Techniques › Formulation
Heat stability (UHT RTD) · Emulsifiers & stabilizers · Sweeteners · Flavor masking · Protein per serving

#### 4.2.9 Quality & Standards › Regulation & Testing
FDA protein claims (good/excellent source) · Supplement vs conventional food · NSF Certified for Sport · Informed Sport

### 4.3 Energy Academy (8 subgenres, 40 styles)

#### 4.3.1 Energy Drinks
Standard energy drink · Sugar-free/zero · Energy shots · Fitness/performance energy · Natural/organic energy

#### 4.3.2 Caffeine Sources
Caffeine anhydrous · Guarana · Green coffee extract · Green tea extract · Guayusa · Kola nut

#### 4.3.3 Pre-Workout
Stimulant pre-workout · Stim-free (pump) pre-workout · Beta-alanine · Creatine monohydrate · L-citrulline

#### 4.3.4 RTD Caffeinated
Canned cold brew · Nitro coffee · RTD tea · Caffeinated sparkling water

#### 4.3.5 Focus & Nootropic
Caffeine + L-theanine · Citicoline/Alpha-GPC drinks · Lion's mane drinks

#### 4.3.6 Science › Caffeine Science
Adenosine receptor antagonism · Half-life (~3–7 h) · ~80 mg per 250 mL can · FDA: ~400 mg/day healthy adults · EFSA: 200 mg single dose · CYP1A2 metabolism

#### 4.3.7 Cultivation & Raw Materials › Other Ingredients
Taurine · B vitamins · Glucuronolactone · Ginseng · L-carnitine · Sugar vs non-nutritive sweeteners · Electrolytes

#### 4.3.8 Quality & Standards › Regulation
Beverage vs supplement labeling · Caffeine disclosure · EU 'high caffeine content' label (>150 mg/L) · Restrictions on sales to minors (some countries)

## 5. Culture Guild — Carbonation, living systems, and what comes next (34 subgenres, 219 styles)

Grouped by living systems, carbonation, place, and shared traditions.

### 5.1 Carbonated Academy (10 subgenres, 61 styles)

#### 5.1.1 Classic Soda Styles
Cola · Lemon-lime · Root beer · Ginger ale (dry/golden) · Cream soda · Orange soda · Grape soda · Pepper-style soda · Birch beer · Sarsaparilla

#### 5.1.2 Mixers & Tonics
Tonic water · Club soda · Bitter lemon · Ginger beer (soft)

#### 5.1.3 Modern Sodas
Prebiotic soda · Probiotic soda · Craft cane-sugar soda · Diet/zero-sugar soda · Italian soda · Sparkling juice

#### 5.1.4 Cordials & Squashes
Squash (UK fruit concentrate) · Cordial · Drinking vinegar (shrub syrup) · Soda syrup

#### 5.1.5 Carbonation Methods
Forced carbonation (CO2) · Natural carbonation (bottle-conditioned) · Nitrogenation (nitro) · Post-mix (fountain) · Pre-mix (bottle/can)

#### 5.1.6 Science › Carbonation Science
CO2 solubility (Henry's law) · Volumes of CO2 · Soda ~2.5–4 volumes · Temperature & pressure · Carbonic acid · Nucleation & bubble retention

#### 5.1.7 Production › Syrup Making
Simple syrup · Flavor concentrates & emulsions · Throw ratio (e.g. 1:5) · Brix targets · Batch mixing · Bag-in-box (post-mix)

#### 5.1.8 Cultivation & Raw Materials › Sweeteners
Sucrose (cane sugar) · HFCS 55 · Aspartame · Sucralose · Acesulfame K · Stevia (Reb M) · Monk fruit · Erythritol · Allulose

#### 5.1.9 Cultivation & Raw Materials › Acids & Preservatives
Phosphoric acid (colas) · Citric acid · Malic acid · Tartaric acid · Sodium benzoate · Potassium sorbate

#### 5.1.10 Production › Packaging
Glass bottle · Aluminum can · PET bottle · Fountain (post-mix) · Keg

### 5.2 Fermented Academy (16 subgenres, 104 styles)

#### 5.2.1 Sake Grades (Tokutei Meishoshu)
Junmai Daiginjo · Daiginjo · Junmai Ginjo · Ginjo · Tokubetsu Junmai · Tokubetsu Honjozo · Junmai · Honjozo

#### 5.2.2 Sake Styles
Futsushu (table sake) · Nigori · Namazake · Genshu · Koshu (aged) · Taruzake · Sparkling sake · Kimoto · Yamahai

#### 5.2.3 Cider
Modern/standard cider · Heritage cider · English cider · French cidre (Normandy/Brittany) · Spanish sidra (Asturias/Basque) · Ice cider (cidre de glace) · Hopped cider

#### 5.2.4 Perry
Modern perry · Traditional perry (perry-pear varieties)

#### 5.2.5 Mead
Traditional mead (dry/semi-sweet/sweet) · Melomel (fruit) · Cyser (apple) · Pyment (grape) · Metheglin (spice) · Braggot (malt) · Sack mead · Hydromel (session)

#### 5.2.6 Barrel Toasting & Aging › Barrel & Wood Aging
Taruzake (Japanese cedar, sugi) · Barrel-aged cider (ex-bourbon/rum) · Normandy/Asturian cider in oak vats · Barrel-aged mead · See Wine Academy: oak, toast, sizes

#### 5.2.7 Fermented Milks
Milk kefir · Kumis (mare's milk) · Drinkable yogurt · Lactobacillus fermented milk drink

#### 5.2.8 Grain & Rice Ferments
Kvass · Boza · Amazake · Makgeolli · Chicha de jora

#### 5.2.9 Plant & Fruit Ferments
Water kefir (tibicos) · Tepache · Pulque · Ginger beer (ginger bug) · Palm wine

#### 5.2.10 Science › Kefir Microbiology
Kefir grains (kefiran matrix) · Lactobacillus kefiranofaciens · Lactococcus · Leuconostoc · Yeasts (Saccharomyces, Kluyveromyces) · Water kefir grains (dextran)

#### 5.2.11 Science › Koji & Aspergillus
Aspergillus oryzae (yellow koji) · Aspergillus luchuensis (black koji, awamori) · Aspergillus sojae · Koji making (seikiku) · Saccharification

#### 5.2.12 Cultivation & Raw Materials › Sake Rice & Polishing
Yamada Nishiki · Omachi · Gohyakumangoku · Miyama Nishiki · Seimaibuai (polishing ratio) · Daiginjo ≤50% · Ginjo ≤60% · Shinpaku (starchy heart)

#### 5.2.13 Production › Sake Brewing Process
Washing & soaking · Steaming · Koji making · Shubo/moto (yeast starter) · Sandan shikomi (3-stage addition) · Moromi (main mash) · Multiple parallel fermentation · Pressing (jōsō) · Filtration · Hi-ire (pasteurization) · Aging

#### 5.2.14 Techniques › Moto (Starter) Methods

Sokujō-moto · Kimoto · Yamahai-moto · Bodaimoto

#### 5.2.15 Production › Cider Making
Sweet apples · Sharp apples · Bittersweet apples · Bittersharp apples · Milling & pressing · Wild vs cultured yeast · Keeving · Malolactic fermentation · Traditional method (cider) · Back-sweetening

#### 5.2.16 Production › Mead Making
Honey varietals · Must preparation · Staggered nutrient additions (SNA) · Yeast selection · Degassing · Aging · Stabilize & back-sweeten

### 5.3 Regional Academy (8 subgenres, 54 styles)

#### 5.3.1 Latin America
Horchata (Mexican rice) · Agua fresca · Chicha morada · Yerba mate · Tereré · Champurrado · Atole · Tejate · Aguapanela

#### 5.3.2 Asia
Bubble tea (boba) · Masala chai · Lassi · Thai iced tea · Vietnamese iced coffee (cà phê sữa đá) · Hong Kong milk tea · Teh tarik · Sikhye · Barley tea (mugicha/boricha)

#### 5.3.3 Middle East & Africa
Ayran · Doogh · Jallab · Sahlab · Moroccan mint tea · Qahwa (Arabic coffee) · Bissap/karkadé (hibiscus) · Sobia · Tamarind drink

#### 5.3.4 Europe
Horchata de chufa (Valencia) · Kompot · Sbiten · Elderflower cordial · Mulled wine (Glühwein)

#### 5.3.5 North America
Egg cream (New York) · Switchel (New England) · Sweet apple cider (unfiltered) · Root beer float · Hot chocolate

#### 5.3.6 Caribbean & Pacific
Sorrel (Jamaica) · Mauby · Sea moss drink · Peanut punch · Kava (Fiji/Vanuatu/Tonga)

#### 5.3.7 Techniques › Preparation: Americas
Mate: gourd & bombilla · Tereré: cold water/juice infusion · Horchata: soaked rice & cinnamon · Chicha de jora: malted corn ferment · Chicha morada: purple corn boil · Atole: masa cooked in water/milk

#### 5.3.8 Techniques › Preparation: Asia & Beyond
Boba: tapioca pearls boiled & sugared · Masala chai: tea simmered with milk & spices · Teh tarik: 'pulled' between vessels · Thai tea: cloth-filter brew · Cà phê sữa đá: phin drip on condensed milk · Horchata de chufa: tiger nuts macerated